

Lecture 1

INTRODUCTION TO PROBABILITY THEORY

18.600

1 - BASIC CONCEPTS

Experiments, outcomes, events & probabilities

EXPERIMENT $\longrightarrow$ OUTCOMES (Ω)

- Rolling a dice $\rightsquigarrow \{1, 2, 3, 4, 5, 6\}$ ($\omega \in \Omega = \{1, \dots, 6\}$)
- Tossing a coin $\rightsquigarrow \{H, T\}$ ($\omega \in \Omega = \{H, T\}$)

The collection of outcomes is called SAMPLE SPACE

$\hookrightarrow$ Quite often is denoted by Ω

A sample space can be $\begin{cases} \text{finite} & (\text{like } \Omega = \{1, \dots, 6\}, \Omega = \{A, B\}, \dots) \leftarrow \\ \text{infinite} & (\text{like } \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \dots) \end{cases}$

$\hookrightarrow$ For the moment, we always assume Ω finite!

A PROBABILITY on a SAMPLE SPACE Ω is a real number between 0 and 1, assigned to all $\omega \in \Omega$, denoted $P(\omega)$, such that

$$\sum_{\omega \in \Omega} P(\omega) = 1.$$

Examples: [Rolling a die with 8 faces]

Here $\Omega = \{1, \dots, 8\}$. A possible probability measure on Ω is the UNIFORM DISTRIBUTION on Ω , that is

$$P(\omega) = \frac{1}{8} = \frac{1}{|\Omega|} \quad \forall \omega \in \Omega$$

↪ $|\Omega| :=$ the cardinality of Ω

Definition: An EVENT is a subset of the sample space Ω . That is, an EVENT is a collection of outcomes.

Example: Let $A = \{\text{The outcome of "rolling a die" is ODD}\}$, then

$$A = \{1, 3, 5\} \quad (\Omega = \{1, \dots, 6\})$$

Definition: The PROBABILITY OF AN EVENT $A \subseteq \Omega$ is equal to

$$P(A) := \sum_{\omega \in A} P(\omega)$$

Example: If $A = \{1, 3, 5\}$, and we have the unif. distrib. on $\Omega = \{1, \dots, 6\}$ the

$$P(A) = P(1) + P(3) + P(5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

3 MORE EXAMPLES:

① Our experiment is now "Rolling two dice with 6 faces". Now we have that

$$\begin{aligned} \Omega &= \{(i, j) \mid 1 \leq i \leq 6, 1 \leq j \leq 6, i, j \in \mathbb{N}\} \\ &= \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), \dots, (6, 6)\} \end{aligned}$$

Q: What is $|\Omega| = ?$ $|\Omega| = 36$ because there are 6 possible values for each coordinate and so $6 \times 6 = 36$ outcomes.

What is a natural choice for P ?

$$P(\omega) = \frac{1}{|\Omega|} = \frac{1}{36}, \quad \forall \omega \in \Omega$$

Now we consider the event

$$\begin{aligned} A &= \{ \text{The sum of the two dice is } 5 \} \\ &= \{ (1, 4), (2, 3), (3, 2), (4, 1) \} \end{aligned}$$

$$\Rightarrow P(A) = 4 \times \frac{1}{36} = \frac{1}{9}$$

□

② Consider the experiment "We toss 3 times a coin":

$$\Omega = \{ HHH, HHT, \dots, TTT \} \quad |\Omega| = 2 \cdot 2 \cdot 2 = 8$$

Again

$$P(\omega) = \frac{1}{8} \quad \forall \omega \in \Omega \quad \left(\text{Note that: } \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \right)$$

IT IS NOT
BLACK MAGIC



$$A = \{ \text{There is at most 1 H} \} = \{ TTT, HTT, TTH, THT \}$$

$$\Rightarrow P(A) = \frac{4}{8} = \frac{1}{2}$$

③ Experiment is "We toss a coin n times", where $n \in \mathbb{N}$, then

$$\Omega = \{ ? \} \quad \text{but} \quad |\Omega| = 2^n$$

and what is a "natural" prob. measure?

$$P(\omega) = \frac{1}{|\Omega|} = \frac{1}{2^n} = 2^{-n}, \quad \forall \omega \in \Omega.$$

Now we fix $k \in \mathbb{N}$, $0 \leq k \leq n$.

$A = \{ \text{Tossing the coin } n \text{ times we get } k \overset{\text{"H"}}{\uparrow} \text{ heads} \}$

$$P(A) = ? = \frac{|A|}{|\Omega|} \quad \text{but } |A| \text{ is a bit of a mess!}$$

The fundamental principle of counting

Suppose we have k tasks to do and the i -th task can be done in n_i ways.

In how many ways can you do the k tasks?

$$\text{ANSWER} = \prod_{i=1}^k n_i = n_1 \cdot n_2 \cdot \dots \cdot n_k.$$

We assume that every task
DO NOT affect any other task.

WE FOCUS ON 3 DIFFERENT SITUATIONS:

- ① We have a deck of cards and we want to put them down on the table one by one. In how many different ways I can do that?

$$\text{ANS} = 54 \cdot 53 \cdot \dots \cdot 1 = 54! \quad (n! = n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1)$$

Def: Given a number $n \in \mathbb{N}$, the FACTORIAL of n is denoted by $n!$ and is equal to $n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1$.

Remark: $n!$ counts the possible ways of reordering n distinct objects.